STUDENT NUMBER:

APSC 171 - Term Test 2
Tuesday November 5, 2024

A. Ableson L. Lu D. Prasad

Last Name (BLOCK LETTERS): _____

First Name: _____

Student #: _____

- This is a 2 hour test.
- No textbooks, notes, or aids other than your calculator (Casio 991 series) are allowed.
- For full marks, you must show all your work and explain how you arrived at your answers, unless explicitly told to do otherwise.
- Wherever appropriate, **include units in your answers**.
- When drawing graphs, **add labels and scales on all axes**.
- Correct final answers with incorrect support may be given no credit.
- While waiting to start, fill in your name and student number above.
- If you need more room, there are blank pages at the end of the test. **If you use these pages, you must provide clear directions on the original page**, e.g. "Continued on page 10".
- Only answers written in this booklet will be graded; all scrap paper from the test will be discarded.

Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Total
6	12	6	4	5	7	6	7	7	60

STUDENT NUMBER:

1. The power used by a motor can be measured in watts; this is also a measure of the **rate** of energy usage, because (1 watt) = 1 Joule/second.

   Below is a table showing the power used by an electric motor driving a small cart at high speed around a long-distance course.

Time (s)	0	15	30	45	60	75	90
Power (W)	45	42	46	52	39	16	0

   (a) Use a right sum with 6 intervals, i.e. RIGHT(6), to estimate the total energy (in Joules) used by the motor during the cart's 90 second run.
   Reminder from the front page: include units in all answers when they are available. [/3]

   (b) Use a trapezoidal sum with 6 intervals, i.e. TRAP(6), to estimate the total energy (in Joules) used by the motor during the cart's 90 second run. [/2]

   (c) In the experiment above, a single fresh battery was used to power the motor, and by the end of the 90 seconds, the battery's energy was all discharged. Based on your energy usage estimate from parts (a) and (b), which battery type, AAA, AA, C or D, was used to power the motor?
   Note that total battery energy is usually reported in mAh (milli-amp hours); AAA, AA, C and D batteries are 1.5 V batteries, and for them 1 mAh = 5.4 J. [/1]

   - AAA: 600-1000 mAh
   - AA: 2000-3000 mAh
   - C: 5000-7000 mAh
   - D: 13000-17000 mAh

STUDENT NUMBER:

2. Evaluate the following definite and indefinite integrals. As always, show your work.

(a) $\int xe^{-2x}\,dx$ [/4]

(b) $\int x^2 \sin(x^3)\,dx$ [/4]

(c) $\int_0^{0.5} \frac{1}{\sqrt{1-x^2}}\,dx$ [/4]

STUDENT NUMBER:

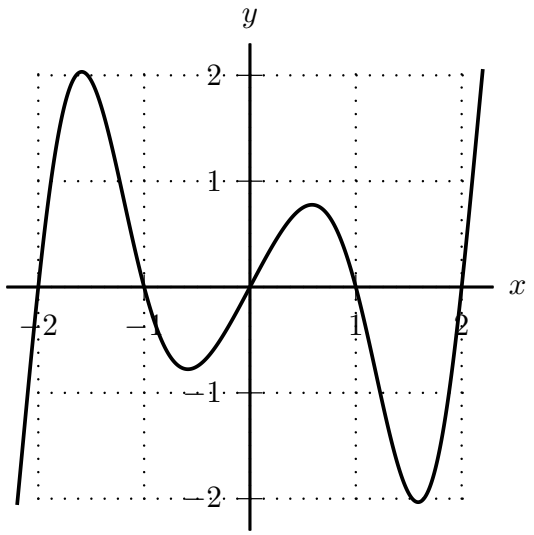
3. Consider the integral $\int x \ln(\sqrt{x})\, dx$.

(a) Evaluate the indefinite integral. [/5]

(b) Determine by differentiating whether or not your answer to part (a) is correct. State your conclusion clearly for the grader. [/1]

STUDENT NUMBER:

4. Consider the function $f(x)$ graphed at right.



We are interested in ranking the value of the following definite integrals.

A. $\int_{-2}^{-1} f(x)\ dx.$

B. $\int_{-1}^{0} f(x)\ dx.$

C. $\int_{-1}^{1} f(x)\ dx.$

D. $\int_{1}^{2} f(x)\ dx.$

Put the capital letters representing the value of the associated integral into the inequality sequence below. For this question no explanation or calculations are necessary.
Note there are 6 slots but only 4 letters/integrals, so two spaces will have to be left blank. [/4]

_____ ≤ _____ ≤ -1 ≤ _____ ≤ _____ ≤ 1 ≤ _____ ≤ _____

---

5. (a) Without using an anti-derivative, determine the **exact** value of the integral $\int_{-4}^{0} (\sqrt{(16-x^2)}-2)\ dx.$
Hint: a sketch will be essential. Decimal answers, e.g. 4.1234, will not be accepted for this question. [/3]

(b) Compute the exact value of the related integral $\int_{-4}^{0} (\mathbf{5}\sqrt{(16-x^2)}-2)\ dx.$ [/2]

STUDENT NUMBER:

6. An LCD monitor has a 4:3 ratio of width to height. The sale price p of a monitor (in dollars) is determined by the length of its diagonal length (in inches):

$$p(d) = 8.5\ d.$$

The manufacturing cost m, however, is related to the area a of the monitor (in square inches):

$$m(a) = \frac{a^2}{8000}.$$

What size of monitor makes the most profit (i.e., maximizes $p - m$)? **As your final answer, report the length of the optimal design's diagonal length** d.
Note: you do **not** need to use the first or second derivative test for this problem to confirm the critical point type. [/7]

Optimal diagonal d: _____

STUDENT NUMBER:

7. A compressed dry sponge is initially 8 cm in length, and 5 cm in both width and height. It is dropped in water and begins to expand. You note that:

   - the width and height remain equal to each other;
   - the length increases at a rate of 3 cm/min; and
   - the volume increases at a rate of 120 cm^3/min.

   (a) At the moment the sponge is dropped in the water, how fast is its width increasing? [/4]

   (b) At what rate is the **surface area** of the sponge increasing at this same moment? [/2]

STUDENT NUMBER:

8. The relation $(x-y)^2+6x-10y+29=0$ defines a parabola.

   (a) Find the (x,y) coordinates of point(s) where the parabola has a horizontal tangent line. [/3]

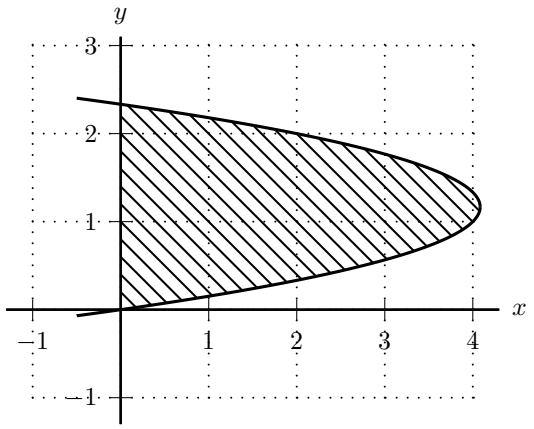
   (b) The point (5,6) lies on the parabola. Find the formula for the tangent line to the parabola at the point (5, 6). [/3]

   (c) Using your answer to part (b), estimate the value of y on the parabola at $x=4.95$. [/1]

STUDENT NUMBER:

9. The following questions are unrelated.

(a) Find the area of shaded region at right, bounded between the vertical axis and the relation $x = 7y - 3y^2$. [/4]



(b) Indicate whether the following statements are true or false by circling the appropriate answer. No explanation is required. [/3]

When computing an indefinite integral like $\int f(x)\,dx$, the $+C$ term in the anti-derivative is optional because it will cancel later in the calculation. T / F

Consider a function $g(x)$ with a critical point $x = a$ where $g'(a) = 0$. The second derivative test indicates whether a critical point is a local max if $g''(a) < 0$, local min if $g''(a) > 0$, or undetermined if $g''(a) = 0$. T / F

Consider a function $f(x)$ that is always increasing and concave down. On any interval, the TRAP sum approximation to the integral of $f(x)$ will always under-estimate the corresponding exact integral value of $f(x)$. T / F

STUDENT NUMBER:

Space for additional work. **Indicate clearly which question you are continuing if you use this space.**

STUDENT NUMBER:

Space for additional work. **Indicate clearly which question you are continuing if you use this space.**

STUDENT NUMBER:

Space for additional work. **Indicate clearly which question you are continuing if you use this space.**